

Problem-Solving and Programming Completed SPARK Reference

YardFresh Community Market - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: YardFresh Community Market

Sector: Business

Questions being answered

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the amount each customer pays after discount and GCT and display the total sales processed.

How SPARK completed it

- State what information the program receives.
- State the main calculation or decision the program performs.
- State the result the program must display.

Completed result

Problem statement: Calculate the amount each customer pays after discount and GCT and display the total sales processed.

2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

How SPARK completed it

- List inputs, processes and outputs before writing pseudocode.
- Use indentation to show the selection and loop structure.
- Keep variable names consistent from the algorithm through to the program.

Completed result

Inputs: Customer name, Purchase subtotal, Number of customers

Processes: Apply 5% discount at or above JMD 10,000, Calculate 15% GCT on the discounted amount, Calculate final amount, Accumulate total sales

Outputs: Customer name, Discount, GCT, Final amount, Total sales

Pseudocode:

START

SET totalSales <- 0

INPUT customerCount

FOR customer <- 1 TO customerCount

INPUT customerName, subtotal

IF subtotal >= 10000 THEN

SET discount <- subtotal * 0.05

ELSE

SET discount <- 0

ENDIF

SET taxable <- subtotal - discount

SET gct <- taxable * 0.15

SET finalAmount <- taxable + gct

SET totalSales <- totalSales + finalAmount

DISPLAY customerName, discount, gct, finalAmount

NEXT customer

DISPLAY totalSales

STOP

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

How SPARK completed it

- Choose values that exercise each branch of the selection statement.
- Follow the pseudocode in order and record important variable values.
- Calculate the expected output before running the program.

Completed result

The completed SPARK trace table includes the project's supplied normal examples and adds a boundary/validation discussion in the completed PDF.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as MarketCheckout_YourFullName.pas.

How SPARK completed it

- Create variables using suitable data types.
- Translate each pseudocode step into program statements in the same logical order.
- Compile or run often so errors are found in small sections.
- Use the same test data from the trace table.

Completed result

The SPARK reference implementation is MarketCheckout_YourFullName.pas. The completed PDF includes the full source code.

2. Test the program and capture evidence that the expected and actual results agree.

How SPARK completed it

- Run normal test data first.
- Run the boundary case.
- Run an invalid or unusual input when the program includes validation.
- Capture screenshots showing both the entered data and the program result.

Completed result

The completed SPARK reference compares expected trace-table outputs with the program output and records the result of each test.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

How SPARK completed it

- Create a cover page and table of contents.

- Keep the section order the same as the task sequence.
- Use readable code formatting and captions for screenshots.
- Export the final document to PDF and check every page.

Completed result

Completed documentation filename: ProgramDocumentation_YourFullName.pdf. The SPARK section PDF demonstrates the required order and evidence.

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Problem definition

Calculate the amount each customer pays after discount and GCT and display the total sales processed.

Input - Process - Output

Inputs	Processes	Outputs
Customer name Purchase subtotal Number of customers	Apply 5% discount at or above JMD 10,000 Calculate 15% GCT on the discounted amount Calculate final amount Accumulate total sales	Customer name Discount GCT Final amount Total sales

Pseudocode

```
START
SET totalSales <- 0
INPUT customerCount
FOR customer <- 1 TO customerCount
    INPUT customerName, subtotal
    IF subtotal >= 10000 THEN
        SET discount <- subtotal * 0.05
    ELSE
        SET discount <- 0
    ENDIF
    SET taxable <- subtotal - discount
    SET gct <- taxable * 0.15
    SET finalAmount <- taxable + gct
    SET totalSales <- totalSales + finalAmount
    DISPLAY customerName, discount, gct, finalAmount
NEXT customer
DISPLAY totalSales
STOP
```

Trace table and test data

Name	Subtotal	Discount	GCT	Final Amount
Marsha Lee	8500	0	1275	9775
Owen Grant	12000	600	1710	13110
Ruth Henry	10000	500	1425	10925

Testing note

The normal data above should be supplemented with a boundary case and an invalid or unusual case where the program includes validation. Expected output should be calculated before the program is run.

Pascal reference implementation

```
program MarketCheckout;
var
  i, customerCount: integer;
  customerName: string;
  subtotal, discount, taxableAmount, gct, finalAmount, totalSales: real;
begin
  totalSales := 0;
  write('Number of customers: '); readln(customerCount);
  for i := 1 to customerCount do
    begin
      write('Customer name: '); readln(customerName);
      write('Subtotal: '); readln(subtotal);
      if subtotal >= 10000 then discount := subtotal * 0.05 else discount := 0;
      taxableAmount := subtotal - discount;
      gct := taxableAmount * 0.15;
      finalAmount := taxableAmount + gct;
      totalSales := totalSales + finalAmount;
      writeln(customerName, ' pays ', finalAmount:0:2);
    end;
  writeln('Total sales: ', totalSales:0:2);
end.
```

Documentation order

- Cover sheet
- Table of contents
- Problem definition
- Algorithm
- Source code
- Trace table and test data
- Program execution screenshots
- Final review

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

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